

The electron gas, and the density as the variable

FYS4480/FYS9480 – Week 42, October 15–16, 2026

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Thursday: the infinite homogeneous electron gas in Hartree-Fock — the one realistic system solved in closed form

Friday: density functional theory, part I — densities, the Hohenberg-Kohn theorems, and a first functional from the gas

Reading: lecture notes chapter 6, section 6.14; chapter 8, sections 8.1–8.4; Szabo-Ostlund, chapter 3.

Code: `BookPrograms/chapter06/hartreefock.py` (electron-gas part), `BookPrograms/chapter08/dft.py`; `notebook week42.ipynb`.

No exercise sessions in weeks 42 and 43: the time is for the first midterm.

Plan for week 42

Thursday, 2×45 min: the electron gas (chapter 6, section 6.14)

- The model: electrons, a uniform background, neutrality; plane waves are the Hartree-Fock orbitals for free
- Three divergent pieces and one cancellation; the interaction in momentum space, $4\pi e^2/q^2$
- The reference energy: $\frac{3}{5}\varepsilon_F$ per particle, direct term gone, exchange left
- The single-particle energy in closed form; $F(x)$, the band width $1 + 0.33 r_s$, the effective mass that goes to zero
- The energy per electron $2.21/r_s^2 - 0.916/r_s$: exchange alone binds a metal, correlation moves the minimum

Friday, 2×45 min: density functional theory, part I (chapter 8, sections 8.1–8.4)

- From $3N$ coordinates to three: the electronic Hamiltonian, and what distinguishes one material from another
- One- and two-body densities; the factorisation for a determinant is the boundary of mean-field theory
- The Hohenberg-Kohn theorems, their four-line proof, and what they do *not* give
- The variational equation $\delta E/\delta n = \mu$; Thomas-Fermi from the electron gas, and why it is not enough

Week 43 continues with the Kohn-Sham construction and the local density approximation. No exercises these two weeks: first midterm.

Where we are

Weeks 38–39: Hartree-Fock

- One determinant, orbitals free; $f_{ai} = 0$ (Brillouin)
- Thouless: every nearby determinant is $e^{\hat{T}_1}|c\rangle$
- Stability: $\hat{M} \geq 0$; Lipkin fails, pairing is stable and useless

Weeks 40–41: matrix product states

- No reference at all; the entanglement is restricted instead
- The SVD at every bond; a sweep of small eigenproblems

This week and next: back to the mean field, twice

- **Thursday**: a system where the Hartree-Fock equations need no iteration — translation invariance hands us the orbitals — and where every quantity follows in closed form. We will see the mean field at its best (a metal is bound) and at its worst ($m^* \rightarrow 0$).
- **Friday**: a different bargain. Not a better wave function, but **no wave function**: the density $n(r)$ as the variable, exactly, by a four-line theorem. The electron gas of Thursday becomes the building block of its first functional.

Thursday, lecture 1: The model and its Hamiltonian

The infinite homogeneous electron gas (section 6.14)

Three assumptions

1. N electrons in a cubic box of side L , volume $\Omega = L^3$, periodic boundary conditions
 2. a uniform, stationary background of positive charge, density Ne/Ω , the only external force
 3. the system as a whole is neutral
- The valence electrons of a simple metal; the cornerstone of the local density approximation (Friday and week 43)
 - Very likely the only realistic interacting system for which Hartree-Fock is solved in closed form
 - Its correlations resist everything but quantum Monte Carlo (Ceperley; Ceperley and Tanatar): the benchmark in 3D and 2D
 - Low density: Wigner crystal, a direct consequence of the long range of $1/r$; high density: a liquid

The infinite range of the Coulomb interaction is the whole story: it makes the pieces diverge, the sum finite, and the mean field singular.

Plane waves, and why no self-consistency is needed

$$\psi_{k\sigma}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}} \xi_{\sigma}, \quad k_i = \frac{2\pi n_i}{L}, \quad n_i = 0, \pm 1, \pm 2, \dots \quad \hat{T} = \sum_{k\sigma} \frac{\hbar^2 k^2}{2m} a_{k\sigma}^{\dagger} a_{k\sigma}$$

- The limit $L \rightarrow \infty$ is taken *after* the expectation values: $\sum_{\mathbf{k}} \rightarrow \Omega(2\pi)^{-3} \int d\mathbf{k}$
- Same structure as the momentum-space Hubbard model of chapter 5: the interaction conserves momentum

Many-body connection

Week 39: a determinant is a Hartree-Fock solution iff $f_{ai} = 0$ for every particle-hole pair. Here $f_{ai} \propto \delta_{\mathbf{k}_a, \mathbf{k}_i}$ by momentum conservation, and $\mathbf{k}_a \neq \mathbf{k}_i$ for a particle and a hole — so $f_{ai} = 0$ *identically*. The filled Fermi sphere is the Hartree-Fock state of *any* translation-invariant interaction, with no iteration. Everything that follows is a calculation *in* the Hartree-Fock basis, not a search for it.

Three pieces, three divergences, one cancellation

$$\hat{H} = \hat{H}_{\text{el}} + \hat{H}_b + \hat{H}_{\text{el}-b}, \quad \hat{H}_{\text{el}} = \sum_i \frac{p_i^2}{2m} + \frac{e^2}{2} \sum_{i \neq j} \frac{e^{-\mu|r_i - r_j|}}{|r_i - r_j|}$$

$$\hat{H}_b = \frac{e^2}{2} \iint dr dr' \frac{n(r)n(r')e^{-\mu|r-r'|}}{|r-r'|}, \quad \hat{H}_{\text{el}-b} = -e^2 \sum_i \int dr \frac{n(r)e^{-\mu|r-x_i|}}{|r-x_i|}, \quad n = \frac{N}{\Omega}$$

- $e^{-\mu r}$ is a convergence device: every integral at finite μ , the limit $\mu \rightarrow 0$ at the end
- Individually: $\hat{H}_b = +\frac{1}{2} e^2 \frac{N^2}{\Omega} \frac{4\pi}{\mu^2}$, $\hat{H}_{\text{el}-b} = -e^2 \frac{N^2}{\Omega} \frac{4\pi}{\mu^2}$, direct term of $\hat{H}_{\text{el}}$:

$$+\frac{1}{2} e^2 \frac{N^2}{\Omega} \frac{4\pi}{\mu^2}$$
- **The three cancel.** What survives is the kinetic energy and the exchange term

$$\hat{H} = \sum_{k\sigma} \frac{\hbar^2 k^2}{2m} a_{k\sigma}^\dagger a_{k\sigma} + \frac{e^2}{2\Omega} \sum_{\sigma_1 \sigma_2} \sum_{q \neq 0, k, p} \frac{4\pi}{q^2} a_{k+q, \sigma_1}^\dagger a_{p-q, \sigma_2}^\dagger a_{p\sigma_2} a_{k\sigma_1}$$

The $q = 0$ term — the one that diverged — is what the background removed.

Pause to think: the cancellation

Pause to think #1

(about 4 min – discuss with your neighbour)

1. Why is the electron-background term $-e^2 N^2 (4\pi/\mu^2)/\Omega$ *twice* the background-background term, with the opposite sign? (Count what interacts with what.)
2. The direct term of the electron-electron repulsion supplies exactly the $+\frac{1}{2}$ that is missing. Which contraction of $\langle \Phi_0 | \hat{V} | \Phi_0 \rangle$ is “direct”, and why does it carry N^2 ?
3. For a short-ranged nuclear force there is no background. What survives in $\langle \Phi_0 | \hat{V} | \Phi_0 \rangle$ then?

Answer to pause #1

1. $\hat{H}_b$ counts each pair of background elements once ($\frac{1}{2} \int \int$), $\hat{H}_{el-b}$ counts every electron against the whole background with no $\frac{1}{2}$ — and the two charge densities are equal and opposite. Hence $-2\times$.
2. The direct contraction pairs $a_{k+q}^\dagger$ with a_k and $a_{p-q}^\dagger$ with a_p , forcing $q = 0$: every occupied state sees every other, N^2 times $\int v(r) dr = 4\pi e^2/\mu^2$, with the $\frac{1}{2}$ of the Hamiltonian in front.
3. Both: $\langle \Phi_0 | \hat{V} | \Phi_0 \rangle = \frac{1}{2\Omega} \left(N^2 \int v dr - N \sum_q \int v e^{-iq \cdot r} dr \right)$, direct minus exchange, is valid for any $v(r)$; only the Coulomb direct term is cancelled, and only because the system is neutral.

The interaction in momentum space

Insert the plane waves into $\hat{V} = \frac{1}{2} \sum \langle pq | \hat{V} | rs \rangle a_p^\dagger a_q^\dagger a_s a_r$ with $p = (k_p, \sigma_p)$ etc.:

$$\langle k_p \sigma_p, k_q \sigma_q | \hat{V} | k_r \sigma_r, k_s \sigma_s \rangle = \frac{\delta_{\sigma_p \sigma_r} \delta_{\sigma_q \sigma_s}}{\Omega^2} \iint e^{-i k_p \cdot r_p} e^{-i k_q \cdot r_q} v(r) e^{i k_r \cdot r_p} e^{i k_s \cdot r_q} dr_p dr_q$$

1. Relative coordinate $r = r_p - r_q$ at fixed r_q : the exponentials separate
2. The r_q integral is a Kronecker delta: $k_p + k_q = k_r + k_s$, momentum conservation
3. What is left is $\frac{1}{\Omega} \int v(r) e^{i(k_r - k_p) \cdot r} dr$ — the Fourier transform at the momentum transfer

$$\hat{V} = \frac{1}{2\Omega} \sum_{\sigma \sigma'} \sum_{k p q} \left[\int v(r) e^{-i q \cdot r} dr \right] a_{k+q, \sigma}^\dagger a_{p-q, \sigma'}^\dagger a_{p \sigma'} a_{k \sigma}, \quad \int \frac{e^{-\mu r}}{r} e^{-i q \cdot r} dr = \frac{4\pi}{\mu^2 + q^2} \xrightarrow{\mu \rightarrow 0} \frac{4\pi}{q^2}$$

The Hubbard interaction of chapter 5 in momentum space, with $4\pi e^2/q^2$ in place of the constant U .
The $1/q^2$ at small q is where all the trouble below comes from.

The reference energy

Kinetic energy of the filled Fermi sphere (spin degeneracy 2, $\sum_{\mathbf{k}} \rightarrow \Omega(2\pi)^{-3} \int$):

$$\langle \Phi_0 | \hat{T} | \Phi_0 \rangle = \frac{2\Omega}{(2\pi)^3} \frac{\hbar^2}{2m} \int_0^{k_F} 4\pi k^4 dk = \frac{\hbar^2 \Omega}{10\pi^2 m} k_F^5, \quad N = \frac{2\Omega}{(2\pi)^3} \frac{4}{3} \pi k_F^3 = \frac{\Omega}{3\pi^2} k_F^3$$

$$\boxed{\frac{\langle \Phi_0 | \hat{T} | \Phi_0 \rangle}{N} = \frac{3}{5} \frac{\hbar^2 k_F^2}{2m} = \frac{3}{5} \varepsilon_F = \frac{\hbar^2 (3\pi^2)^{5/3}}{10\pi^2 m} \rho^{2/3}, \quad \rho = \frac{N}{\Omega}, \quad k_F = (3\pi^2 \rho)^{1/3}}$$

- Interaction: Wick on $a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{p}-\mathbf{q}}^\dagger a_{\mathbf{p}} a_{\mathbf{k}}$ leaves two complete contractions — direct ($\mathbf{q} = 0$) and exchange ($\mathbf{k} + \mathbf{q} = \mathbf{p}$, $\sigma = \sigma'$)
- Direct: cancelled by the background. Exchange: $-\frac{N}{2\Omega} \sum_{\mathbf{q}} \frac{4\pi e^2}{q^2}$ restricted to $\mathbf{k}, \mathbf{k} + \mathbf{q}$ both inside the sphere

Density functional connection

Kinetic energy per volume $t_0 \propto \rho^{5/3}$ and exchange energy per particle $\propto \rho^{1/3}$, as functions of *the density alone*: Friday's Thomas-Fermi functional and week 43's local density approximation.

Pause to think: direct and exchange

Pause to think #2

(about 3 min – discuss with your neighbour)

Write the interaction as $\frac{1}{2} \sum v(q) a_{k+q,\sigma}^\dagger a_{p-q,\sigma'}^\dagger a_{p\sigma'} a_{k\sigma}$ and take the expectation value in the Fermi sea with Wick's theorem (weeks 36–37).

1. Draw the two complete contraction patterns. Which conditions on q , k , p , σ , σ' does each impose?
2. Which one carries a minus sign, and why?
3. Why does the exchange term not need the background to be finite?

Answer to pause #2

1. Direct: $(a_{k+q}^\dagger, a_k)(a_{p-q}^\dagger, a_p)$, nested lines, needs $q = 0$, any spins. Exchange: $(a_{k+q}^\dagger, a_p)(a_{p-q}^\dagger, a_k)$, crossed lines, needs $p = k + q$ and $\sigma = \sigma'$ (the operators carry the spin labels).
2. Exchange: one crossing of contraction lines, $(-1)^1$. Same rule as for the stability matrix of week 40.
3. Because $q \neq 0$ there: $\sum_q 4\pi e^2/q^2$ over the Fermi sphere is a convergent integral in three dimensions, $\int d^3q/q^2 \sim \int dq$. The divergence sat entirely in the $q = 0$ Fourier component, which is the uniform (direct) part.

Thursday, lecture 2: Single-particle energies and the energy per electron

The Hartree-Fock single-particle energy

Kinetic energy plus the exchange self-energy, the sum running over the occupied states:

$$\varepsilon_k^{\text{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2}{\Omega^2} \sum_{p \leq k_F} \int d\mathbf{r} e^{i(\mathbf{p}-\mathbf{k}) \cdot \mathbf{r}} \int d\mathbf{r}' \frac{e^{i(\mathbf{k}-\mathbf{p}) \cdot \mathbf{r}'}}{|\mathbf{r} - \mathbf{r}'|}$$

$$\boxed{\varepsilon_k^{\text{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2 k_F}{2\pi} \left[2 + \frac{k_F^2 - k^2}{kk_F} \ln \left| \frac{k + k_F}{k - k_F} \right| \right]}$$

- Week 39: $\varepsilon_p = \langle p | \hat{h}_0 | p \rangle + \sum_{i \leq F} \langle pi | \hat{v} | pi \rangle_{\text{AS}}$; the direct part of the sum is again cancelled by the background, the exchange part is what is written
- Every state is lowered; the bottom of the band far more than the top
- The logarithm is harmless at $k = k_F$ (it is multiplied by zero) but its *derivative* is not

The derivation in four steps

1. Convergence factor $e^{-\mu|r-r'|}$, $\sum_{\mathbf{p}} \rightarrow \Omega(2\pi)^{-3} \int d\mathbf{p}$; change to $x = r - r'$, $y = r'$: the y integral gives Ω
2. The Fourier transform of the screened Coulomb potential:

$$\int d\mathbf{x} e^{i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}} \frac{e^{-\mu x}}{x} = 4\pi \int_0^\infty \frac{\sin(|\mathbf{p}-\mathbf{k}|x)}{|\mathbf{p}-\mathbf{k}|} e^{-\mu x} dx = \frac{4\pi}{\mu^2 + |\mathbf{p}-\mathbf{k}|^2}$$

and the limit $\mu \rightarrow 0$ is now harmless

3. Angular integral with $u = \cos \theta$:

$$\frac{e^2}{2\pi^2} \int_{p \leq k_F} \frac{d\mathbf{p}}{|\mathbf{p}-\mathbf{k}|^2} = \frac{e^2}{\pi} \int_0^{k_F} p^2 dp \int_{-1}^1 \frac{du}{p^2 + k^2 - 2pk u}, \quad \int_{-1}^1 \frac{du}{p^2 + k^2 - 2pk u} = \frac{1}{pk} \ln \left| \frac{p+k}{p-k} \right|$$

4. Radial integral by $x = p + k$, $y = p - k$:

$$\frac{e^2}{k\pi} \int_0^{k_F} p \ln \left| \frac{p+k}{p-k} \right| dp = \frac{e^2}{k\pi} \left[kk_F + \frac{k_F^2 - k^2}{2} \ln \left| \frac{k+k_F}{k-k_F} \right| \right]$$

Rearranging gives the boxed result. The notebook checks step 4 by numerical quadrature at $k = 0.3k_F$ and $k = 1.7k_F$.

Dimensionless form: r_s and $F(x)$

$$\rho = \frac{k_F^3}{3\pi^2} = \frac{3}{4\pi r_0^3} \Rightarrow k_F = \frac{(9\pi/4)^{1/3}}{r_0} = \frac{1.9192}{r_0}, \quad r_s = \frac{r_0}{a_0}, \quad a_0 = \frac{\hbar^2}{me^2}, \quad x = \frac{k}{k_F}$$

$$F(x) = \frac{1}{2} + \frac{1-x^2}{4x} \ln \left| \frac{1+x}{1-x} \right|, \quad \boxed{\frac{\varepsilon_k^{\text{HF}}}{\varepsilon_F^0} = x^2 - \frac{4}{\pi k_F a_0} F(x) = x^2 - 0.663 r_s F(x)}$$

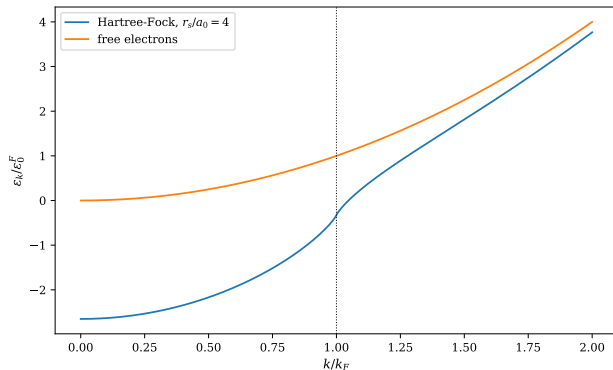
with $\varepsilon_F^0 = \hbar^2 k_F^2 / 2m$ the free Fermi energy; r_s lies between about 2 and 6 for most metals.

x	$F(x)$	$\varepsilon_k^{\text{HF}} / \varepsilon_F^0$	x^2
0.00	1.000000	-2.652000	0.0000
0.50	0.911980	-2.168570	0.2500
0.75	0.783779	-1.516081	0.5625
1.00	0.500000	-0.326000	1.0000
1.25	0.252812	0.892042	1.5625
2.00	0.088020	3.766570	4.0000

$r_s = 4$ (hartreefock.py, Table 6.1)

- $F(0) = 1$, $F(1) = \frac{1}{2}$, $F \rightarrow 0$ like $1/(3x^2)$ for large x
- Smooth everywhere except $x = 1$: finite value, **infinite slope**
- One number, $0.663 r_s$, sets the size of the whole effect: 1.3 to 4.0 at metallic $r_s = 2-6$. Exchange is never a small perturbation

The band at $r_s = 4$ (chapter 6, Fig. 6.5)



- The shift is *not* constant: -2.652 at the bottom, -1.326 at k_F , -0.233 at $x = 2$. A state deep in the sea exchanges with every occupied state; a state at k_F with fewer
- A rigid shift would only move the zero of energy. The k -dependence is the physics
- The occupied band lies below zero out to $x \simeq 1.05$: at k_F exchange (-1.326) beats kinetic energy ($+1$) — **this is what binds the gas**
- The kink at $x = 1$: not a plotting artefact but $F'(1) = -\infty$

Pause to think: the band width

Pause to think #3

(about 4 min – discuss with your neighbour)

1. Using $F(0) = 1$ and $F(1) = \frac{1}{2}$, find the width of the occupied band, $\varepsilon_{k_F}^{\text{HF}} - \varepsilon_0^{\text{HF}}$, in units of ε_F^0 and as a function of r_s . Check it against the table at $r_s = 4$.
2. The free gas has width exactly 1. Is the Hartree-Fock correction of the right sign? Of the right size? (Sodium has $r_s = 3.93$ and a measured band width close to, if anything slightly below, the free-electron value.)
3. Where does the error come from, physically?

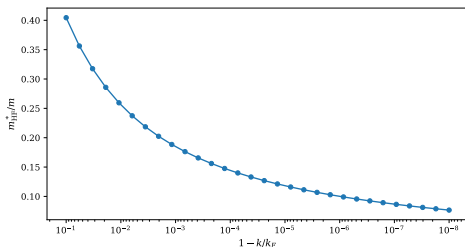
Answer to pause #3

1. $\Delta\epsilon^{\text{HF}} = [1 - 0.663r_s \cdot \frac{1}{2}] - [0 - 0.663r_s \cdot 1] = 1 + 0.3315 r_s$; at $r_s = 4$:
 $2.326 = -0.326 - (-2.652)$. ✓
2. Right sign — exchange is larger for the deep states, so it stretches the band — but far too large: more than doubled at sodium's density, where experiment finds roughly the free value.
3. The bare $1/q^2$ exchange is unscreened. In the metal the other electrons rearrange and the effective interaction decays beyond the screening length; the small- q part of the exchange, which is precisely what lowers the deep states most, is removed. Correlation (RPA, chapter 7) undoes most of the stretching.

The effective mass goes to zero — logarithmically

Expand about k_F and compare with $\varepsilon_k^{(0)} = \varepsilon_F^0 + (\hbar^2 k_F/m)(k - k_F) + \dots$:

$$m_{\text{HF}}^* \equiv \hbar^2 k_F \left(\frac{\partial \varepsilon_k^{\text{HF}}}{\partial k} \right)_{k_F}^{-1} = \frac{2m}{2 - 0.663 r_s F'(1)}, \quad F'(x) \rightarrow -\infty \text{ like } \ln|1-x| \text{ as } x \rightarrow 1$$



Measured

$r_s = 4$, approaching from below: $m^*/m = 0.256, 0.185, 0.144, 0.118, 0.100$ at $1 - x = 10^{-2}, \dots, 10^{-6}$; 0.076 at 10^{-8} . Eight decades cost a factor of five: $m^* \sim 1/\ln(1-x)$.

- Level density $n(\varepsilon) = \frac{\Omega k^2}{2\pi^2} \left(\frac{\partial \varepsilon}{\partial k} \right)^{-1} \rightarrow 0$ at the Fermi surface
- A finite k-mesh, spacing $\sim 1/L$, so $1 - x \sim N^{-1/3}$, sees a finite, plausible, **basis-dependent** $m^* \approx 0.3$ –0.4. Only the analytic result reveals that there is no limit

Qualitatively wrong, and why

- Metals have a finite density of states at the Fermi level: the linear specific heat and the Pauli susceptibility depend on it. Hartree-Fock gives zero
- The culprit is the $1/q^2$ singularity of the *bare* Coulomb interaction at small momentum transfer, treated without any screening
- In the real metal the other electrons screen it; beyond the screening length the effective interaction falls off exponentially and $F'(1)$ is finite

Many-body connection

Summing the diagrams that produce the screening is exactly what the random-phase approximation of chapter 7 does: the ring diagrams resum the $1/q^2$ into $4\pi e^2/(q^2 + k_{\text{TF}}^2)$ and restore a finite effective mass. The electron gas shows both what a mean field does well and precisely where it must fail — and it points at the next method.

A trap to avoid

A finite-basis Hartree-Fock calculation of the electron gas returns a finite m^* that depends on N . Converging it in N does not converge it to anything: it creeps to zero like $1/\ln N^{1/3}$. Always ask whether the quantity you compute *has* a thermodynamic limit.

The energy per electron

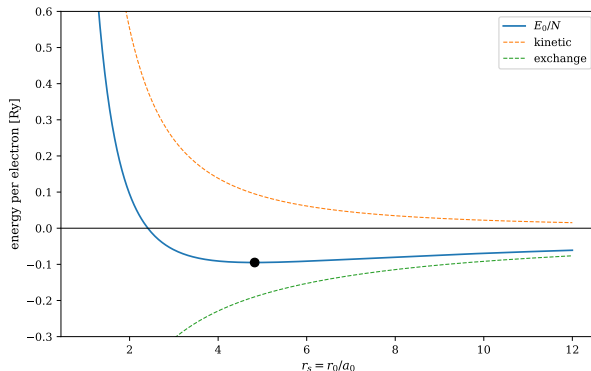
In Rydberg, $e^2/2a_0 = 13.606$ eV, kinetic energy plus exchange:

$$\frac{E_0}{N} = \frac{e^2}{2a_0} \left[\frac{2.21}{r_s^2} - \frac{0.916}{r_s} \right], \quad \frac{3}{5} \left(\frac{9\pi}{4} \right)^{2/3} = 2.2099, \quad \frac{3}{2\pi} \left(\frac{9\pi}{4} \right)^{1/3} = 0.9163$$

r_s	E_0/N [Ry]
1.00	1.294000
2.00	0.094500
3.00	-0.059778
4.00	-0.090875
4.83	-0.094916
6.00	-0.091278
8.00	-0.079969

- Kinetic: $\rho^{2/3} \propto r_s^{-2}$, wins at high density; exchange: $\rho^{1/3} \propto r_s^{-1}$, wins at low density
- Opposite signs, different powers $\Rightarrow$ a minimum:
 $\frac{d}{dr_s} = 0$ at $r_s = \frac{2 \times 2.21}{0.916} = 4.83$,
 $E_0/N = -0.0949$ Ry = -1.29 eV
- Crosses zero at $r_s = 2.41$; the well is **very shallow**: within 5% of the minimum for $4 \leq r_s \leq 6$

Exchange binds a metal (chapter 6, Fig. 6.7)



- A real success: exchange alone, with no correlation whatsoever, explains why a metal is stable against expansion, at a density squarely in the alkali range ($r_s = 3.93$ for Na, 5.62 for Cs)
- The flatness is physics: a system this soft against compression has a small bulk modulus — the alkalis are among the most compressible solids
- And a warning: the correlation energy, about -0.06 Ry near equilibrium, is comparable to the *whole* Hartree-Fock binding. Adding the Ceperley-Alder value (notebook) roughly doubles the binding, to -0.155 Ry, and moves the minimum to $r_s \simeq 4.2$ — same mechanism, different numbers

Pause to think: what a mean field can and cannot do here

Pause to think #4

(about 4 min – discuss with your neighbour)

1. In the Hubbard model of week 38 the plane-wave determinant was stationary but not always minimal. Can the plane-wave state of the electron gas be unstable in the sense of week 39? What would the lower determinant look like?
2. The binding energy comes out too small by roughly a factor of two. Where in the methods map does the missing piece live, and which of our approximations can reach it for this system?

Answer to pause #4

1. Yes. Overhauser showed that the paramagnetic Fermi sphere is *never* the Hartree-Fock minimum of the Coulomb gas: a spin- or charge-density wave, a determinant of Bloch-like orbitals that breaks translation invariance, always lies lower, and at low density the fully polarised (ferromagnetic) sea beats the unpolarised one already at the level of exchange. Stationarity by symmetry is not minimality — the lesson of the Hubbard pause in week 38.
2. The correlation energy: everything beyond one determinant. Second-order perturbation theory *diverges* for the Coulomb gas (the $1/q^4$ from two bare interactions is not integrable at small q); only an infinite resummation of the ring diagrams — the RPA of chapter 7, Gell-Mann and Brueckner — gives the finite $\ln r_s$ term. The electron gas is where “which diagrams, and all of them” was first learned.

What Thursday leaves on the table for Friday

quantity	electron gas, exactly	as a function of
kinetic energy per volume	$t_0 = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} \rho^{5/3}$	ρ alone
exchange energy per electron	$-\frac{0.916}{r_s} \text{ Ry} = -\frac{3e^2}{4\pi} (3\pi^2 \rho)^{1/3}$	ρ alone
correlation energy per electron	from quantum Monte Carlo (Ceperley-Alder)	ρ alone

- Three energies that depend on *nothing but the density*. For a system whose density varies slowly, pretend every small region is a uniform gas at the local $\rho(r)$ and integrate
- That is the oldest idea in density functional theory (Thomas, Fermi, Dirac, 1927–30), and Friday starts by asking whether it can be made exact

Friday: no wave function at all. The density is the variable — by a theorem, not by an approximation.

Friday, lecture 3: From the wave function to the density

A different bargain (chapter 8, introduction)

- Every method so far computes a *wave function*: FCI stores its coefficients, Hartree-Fock optimises its orbitals, RPA the fluctuations around them, the DMRG its tensors. All live in a space of dimension exponential in N
- Density functional theory replaces $\Psi(r_1, \dots, r_N)$, a function of $3N$ coordinates, by the density $n(r)$, a function of **three**
- This is *not* an approximation. Hohenberg and Kohn (1964): the ground-state density determines the external potential, hence the Hamiltonian, hence everything
- The catch, to be stated precisely: the theorem asserts the *existence* of a universal functional and says nothing about its form

Three parts, this week and next

- (i) the tools: the electronic Hamiltonian and the reduced density matrices in coordinate space;
- (ii) the Hohenberg-Kohn theorems and what they do and do not deliver; (iii) the Kohn-Sham scheme, with the functional built from the electron gas (week 43).

Most crystalline materials and a very large fraction of molecular structures have been described with it; Kohn's Nobel Prize in Chemistry, 1998.

The electronic Hamiltonian (section 8.1)

Nuclei and electrons, nothing else needed in principle, nothing soluble in practice. Born-Oppenheimer: the nuclei are ≥ 1836 times heavier, so freeze them:

$$\left[\sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 + v_{\text{ext}}(r_i) \right) + \sum_{i < j} v(r_{ij}) \right] \Psi(r) = E_0 \Psi(r), \quad v_{\text{ext}}(r) = - \sum_j \frac{e^2 Z_j}{|r - R_j|}$$

- The Hamiltonian of chapter 4 written in coordinate space; a generic $v(r_{ij})$, so that nuclear forces are included (with $v_{\text{ext}} = 0$: self-bound)
- **The one thing that matters below:** $\hat{T}$ and $\hat{V}$ are the *same for every system*. The only thing that distinguishes a crystal from a molecule from a cell is $v_{\text{ext}}(r)$

Many-body connection

In chapters 5–7 the single-particle basis carried the external field inside $\hat{h}_0$ and the many-body problem was a matrix in that basis. Here the external field is a *function of one variable*, $v_{\text{ext}}(r)$, and the question becomes: how much of the answer is a functional of a function of one variable?

The one-body density (section 8.2)

$$\hat{n}(r) = \sum_{i=1}^N \delta(r - r_i), \quad n(r) = \frac{\langle \Psi | \hat{n}(r) | \Psi \rangle}{\langle \Psi | \Psi \rangle} = N \int d^3 r_2 \cdots d^3 r_N |\Psi(r, r_2, \dots, r_N)|^2$$

One determinant

$n(r) = \sum_{i=1}^N |\psi_i(r)|^2$ — used in the SCF of week 38

Any state, second quantization

$\hat{n}(r) = \sum_{\alpha\beta} \psi_{\alpha}^*(r) \psi_{\beta}(r) a_{\alpha}^{\dagger} a_{\beta}$; for an FCI state the expectation value needs $\langle \Phi_i | a_{\alpha}^{\dagger} a_{\beta} | \Phi_j \rangle$ — the one-body density matrix

Off the diagonal, with $x = (r, \sigma)$:

$$\rho^{(1)}(x'; x) = N \int dx_2 \cdots dx_N \Psi^*(x', x_2, \dots) \Psi(x, x_2, \dots), \quad n(x) = \rho^{(1)}(x; x), \quad \int n = N$$

- Hermitian; its eigenvalues are the occupation numbers of chapter 1 (natural orbitals), $0 \leq n_k \leq 1$
- Exactly N ones and the rest zeros iff Ψ is a single determinant: then $\rho^{(1)}$ is the projector on the occupied space

The two-body density, and the energy

$$\rho^{(2)}(x_1, x_2; y_1, y_2) = \frac{N(N-1)}{2} \int dx_3 \cdots dx_N \Psi^*(y_1, y_2, x_3, \dots) \Psi(x_1, x_2, x_3, \dots)$$

- The prefactor counts the distinct pairs; the diagonal $\rho^{(2)}(x_1, x_2)$ is the joint probability of a particle at x_1 and another at x_2
- Second quantization: $\hat{\rho}^{(2)}(r_1, r_2) = \sum \rho_{\alpha\gamma}(r_1) \rho_{\beta\delta}(r_2) a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}$ with $\rho_{\alpha\beta}(r) = \psi_{\alpha}^*(r) \psi_{\beta}(r)$

The energy needs only $\rho^{(1)}$ and $\rho^{(2)}$

$$E = \int dx \left[-\frac{\hbar^2}{2m} \nabla_x^2 \rho^{(1)}(x'; x) \right]_{x'=x} + \int dx v_{\text{ext}}(x) n(x) + \int dx_1 dx_2 v(r_{12}) \rho^{(2)}(x_1, x_2)$$

The full wave function is superfluous for the energy. *That* is the origin of every density-based method.

Antisymmetry, and when the pair density factorises

$\Psi(\dots, x_i, \dots, x_j, \dots) = -\Psi(\dots, x_j, \dots, x_i, \dots) \Rightarrow \rho^{(2)}(x, x) = 0$: the Pauli principle as a statement about $\rho^{(2)}$. For a single determinant, Wick's theorem on $\langle \Phi_0 | a^\dagger a^\dagger a a | \Phi_0 \rangle$ leaves exactly two contractions:

$$\rho^{(2)}(x_1, x_2; y_1, y_2) = \rho^{(1)}(x_1; y_1)\rho^{(1)}(x_2; y_2) - \rho^{(1)}(x_1; y_2)\rho^{(1)}(x_2; y_1)$$

diagonal: $\rho^{(2)}(x_1, x_2) = n(x_1)n(x_2) - |\rho^{(1)}(x_1; x_2)|^2, \quad \rho^{(2)}(x, x) = n^2 - n^2 = 0.$

- First term: the naive product, independent particles. Second: exchange, which enforces antisymmetry — the same direct-minus-exchange as Thursday's $\langle \Phi_0 | \hat{V} | \Phi_0 \rangle$
- **This is the boundary of mean-field theory.** A Hartree product would give $n(x_1)n(x_2)$ but is not a fermionic state; a determinant gives the formula above *exactly*; for any correlated state $\rho^{(2)}$ cannot be reduced to $\rho^{(1)}$ at all — the difference is what this whole course calls correlation

Pause to think: where correlation lives

Pause to think #5

(about 4 min – discuss with your neighbour)

1. For two electrons of opposite spin in the same spatial orbital ϕ (helium in Hartree-Fock), write $\rho^{(2)}(r_1\uparrow, r_2\downarrow)$. Does the exchange term contribute? What is the probability of finding them at the same point?
2. Same two electrons with parallel spins in orbitals ϕ_1, ϕ_2 . Now what is $\rho^{(2)}(r, r)$?
3. The exact helium ground state has $\rho^{(2)}(r_1\uparrow, r_2\downarrow)$ *smaller* than $n(r_1)n(r_2)/4$ at short distance. Which of our methods can produce that, and which cannot?

Answer to pause #5

1. $\rho^{(1)}(r_1 \uparrow; r_2 \downarrow) = 0$ (different spins), so $\rho^{(2)} = n_{\uparrow}(r_1)n_{\downarrow}(r_2) = |\phi(r_1)|^2|\phi(r_2)|^2$: a pure product. The two electrons are completely uncorrelated and sit on top of each other with full probability. The *exchange hole* exists only between like spins.
2. $\rho^{(2)}(r, r) = n(r)^2 - |\rho^{(1)}(r; r)|^2 = 0$: a like-spin pair never coincides, whatever the orbitals.
3. The *correlation hole* between unlike spins needs $\rho^{(2)}$ beyond the factorised form: CI, coupled cluster, the DMRG, quantum Monte Carlo — any method with more than one determinant. Hartree-Fock cannot, by construction; density functional theory can, *if* the functional knows about it. That is what E_{xc} is for.

Two faces of one problem

Reduced-density-matrix methods

Determine $\rho^{(2)}$ directly — it is all the energy needs. The obstacle: the **N -representability problem**. Not every Hermitian, antisymmetric, correctly normalised $\rho^{(2)}$ comes from an N -fermion wave function, and characterising those that do is extremely hard.

Density functional theory

Go all the way down to $n(r)$. The obstacle: the **unknown functional**. Everything the wave function knew is now hidden in $F[n]$.

An unknown functional here, an unknown representability condition there — neither has been solved, and they are the same difficulty seen from two sides.

Many-body connection

Chapter 1: the eigenvalues of $\rho^{(1)}$ are the natural occupations, all 0 or 1 exactly for a determinant; their distance from $\{0, 1\}$ measures correlation. That is a property of $\rho^{(1)}$, not of $n(r)$ — the density has thrown it away, and the functional must put it back.

Friday, lecture 4: The Hohenberg-Kohn theorems

The two theorems (section 8.3)

Consider the family $\hat{H} = \hat{T} + \hat{V}_{\text{ext}} + \hat{V}$ with $\hat{T}$, $\hat{V}$ fixed once and for all — properties of electrons, not of materials — and only v_{ext} varying. Non-degenerate ground states, for now.

Hohenberg-Kohn I

The external potential $v_{\text{ext}}(\mathbf{r})$ is determined, up to an additive constant, by the ground-state density $n_0(\mathbf{r})$.

Hohenberg-Kohn II

There is a universal functional $F[n] = T[n] + V_{\text{int}}[n]$, the same for every v_{ext} , such that

$$E[n] = F[n] + \int d^3r v_{\text{ext}}(\mathbf{r})n(\mathbf{r})$$

is minimised, over densities integrating to N , by the true ground-state density, and its minimum is E_0 .

Three variables determine the potential, hence $\hat{H}$, hence Ψ and every ground-state observable. A quantity measured routinely by X-ray diffraction contains, in principle, all of the $3N$ -dimensional Ψ .

The proof of the first theorem: four lines

Suppose $v^{(1)}$ and $v^{(2)}$, differing by more than a constant, gave the *same* ground-state density $n(r)$. They have different $\hat{H}^{(1)}, \hat{H}^{(2)}$ and (unique continuation) different ground states $\Psi^{(1)} \neq \Psi^{(2)}$, energies $E^{(1)}, E^{(2)}$. Use $\Psi^{(2)}$ as a trial state for $\hat{H}^{(1)}$; the variational principle is **strict** for a non-degenerate ground state:

$$E^{(1)} < \langle \Psi^{(2)} | \hat{H}^{(1)} | \Psi^{(2)} \rangle = \langle \Psi^{(2)} | \hat{H}^{(2)} | \Psi^{(2)} \rangle + \int [v^{(1)} - v^{(2)}] n = E^{(2)} + \int [v^{(1)} - v^{(2)}] n$$

Interchange the labels:

$$E^{(2)} < E^{(1)} + \int [v^{(2)} - v^{(1)}] n$$

Add: the integrals cancel and

$$E^{(1)} + E^{(2)} < E^{(1)} + E^{(2)}.$$

- The middle step used *only* that the Hamiltonians differ by a one-body potential and that the density is the same by hypothesis
- Hence no two potentials differing by more than a constant share a ground-state density.
 $n_0 \mapsto v_{\text{ext}}$ is injective

The second theorem follows, and the variational equation

Since n_0 fixes v_{ext} , it fixes $\hat{H}$ and therefore $\Psi[n]$, so

$$F_{\text{HK}}[n] \equiv \langle \Psi[n] | \hat{T} + \hat{V} | \Psi[n] \rangle$$

is a functional of the density alone, and universal because $\hat{T}$, $\hat{V}$ contain no reference to v_{ext} . Rayleigh-Ritz within the family: $E_0 = \min_n E[n]$.

Minimise with $\int n = N$ fixed (section 8.3.4)

$$\frac{\delta}{\delta n(r)} \left\{ E[n] - \mu \left(\int n \, d^3r - N \right) \right\} = 0 \quad \implies \quad \left. \frac{\delta E[n]}{\delta n(r)} \right|_{n_0} = \mu$$

The functional derivative of the energy is constant throughout the system: that constant is the chemical potential, the Fermi level.

Many-body connection

Week 39: the multiplier enforcing normalisation is the energy conjugate to what it constrains — ε_i for $\langle \phi_i | \phi_i \rangle = 1$ there, μ for $\int n = N$ here.

Pause to think: what the proof uses

Pause to think #6

(about 4 min – discuss with your neighbour)

1. Where exactly is non-degeneracy used? What goes wrong in the four lines if $\hat{H}^{(1)}$ has a degenerate ground state?
2. Where is it used that $\psi^{(1)} \neq \psi^{(2)}$? Could two different potentials have the same ground-state *wave function*?
3. The theorem is about *ground-state* densities. Is every reasonable positive function with $\int n = N$ the ground-state density of some v_{ext} ?

Answer to pause #6

1. In the strict inequality $E^{(1)} < \langle \Psi^{(2)} | \hat{H}^{(1)} | \Psi^{(2)} \rangle$. With degeneracy, $\Psi^{(2)}$ might be another ground state of $\hat{H}^{(1)}$, the inequality becomes $\leq$, and the contradiction evaporates. Levy's constrained search (next frame) repairs this.
2. Same place: strictness needs a trial state that is *not* the ground state. If $\Psi^{(1)} = \Psi^{(2)}$ then $\sum_i [v^{(1)} - v^{(2)}](r_i) \Psi = (E^{(1)} - E^{(2)}) \Psi$, so $v^{(1)} - v^{(2)}$ is constant wherever $\Psi \neq 0$ — and unique continuation says that is almost everywhere.
3. No. That is the v -representability problem: the set of ground-state densities is not characterised, and some reasonable densities are not in it. The theorem says nothing about those.

What the theorems do not give (section 8.3.3)

They are existence theorems

Nothing constructs $F[n]$, and nothing suggests it is simple. All the many-body difficulty of chapters 5–11 is still there, moved inside $F[n]$. The practical history of the subject is a history of **approximate functionals**.

v -representability, and the Levy-Lieb repair

$$F_{\text{LL}}[n] = \min_{\Psi \rightarrow n} \langle \Psi | \hat{T} + \hat{V} | \Psi \rangle$$

Search over all antisymmetric Ψ yielding the density n : defined for every N -representable density, equal to F_{HK} where that exists, extends to degenerate ground states — and is, of course, no more computable.

Ground states only

E_0 and n_0 , nothing else. Excitation energies (chapter 7) need a separate construction (time-dependent DFT). The Kohn-Sham eigenvalues of next week are **not** excitation energies.

Building a functional from the electron gas (section 8.4)

Take the one system we solved exactly on Thursday and use it *locally*:

$$t_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} n^{5/3} \quad \Longrightarrow \quad \mathcal{T}_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} \int d^3r (n(r))^{5/3} \quad (\text{Thomas-Fermi})$$

With the classical electrostatic (Hartree) energy $\mathcal{U}_H[n] = \frac{e^2}{2} \iint d^3r d^3r' n(r)n(r')/|r-r'|$,

$$\mathcal{E}_{\text{HK}}[n] = \mathcal{T}_0[n] + \int d^3r v_{\text{ext}} n + \mathcal{U}_H[n] + E_{\text{xc}}[n]$$

- The decomposition is a *definition*, not an approximation: whatever is missing is in E_{xc} by construction
- It is useful only if that remainder is small — **the whole gamble of the subject**
- Thursday's exchange energy, $-\frac{3e^2}{4\pi}(3\pi^2n)^{1/3}$ per electron, integrated the same way, is Dirac's exchange functional, the first piece of E_{xc}

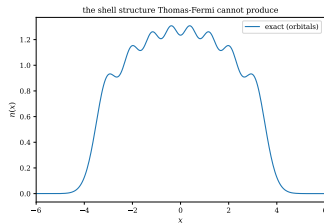
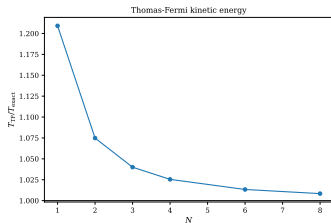
Why Thomas-Fermi is not enough

No orbitals: minimising $\mathcal{E}_{\text{HK}}$ over n gives an *algebraic* equation, enormously cheaper than anything in this course. It also fails, instructively (dft.py: N non-interacting fermions in a 1D trap):

N	T exact	T TF	ratio
1	0.25	0.3023	1.2092
2	1.00	1.0748	1.0748
4	4.00	4.1018	1.0255
8	16.00	16.1343	1.0084

- The ratio improves as promised (smoother density on the scale of the local Fermi wavelength) — but the *absolute* error is 0.05, 0.08, 0.10, 0.13: nearly constant, divided by a growing denominator
- A tenth of a Hartree in the *largest* term of the energy is not available when chemical accuracy is wanted
- Worse: no shell structure at all. Teller: within the model no molecule is bound

The failure that no coefficient can repair (chapter 8, Fig. 8.6)



- Left: the ratio, falling steeply then flattening into a long 0.8% tail
- Right: the exact density of eight particles carries *eight maxima* — the nodal structure of the orbitals surviving the sum — and ends at the turning point $\sqrt{15} \simeq 3.87$ of the highest level. Thomas-Fermi gives a smooth dome
- The shell effects that organise all of atomic and nuclear structure are lost with it

The offending quantity is the kinetic energy; the offending feature is shell structure. Both can be recovered exactly by putting the orbitals back in for that one term. That is the Kohn-Sham construction — next week.

Summary

Summary of week 42

Thursday: the electron gas

- Plane waves are Hartree-Fock for free: $f_{ai} = 0$ by momentum conservation
- Three μ^{-2} divergences cancel; the $q = 0$ term is the background
- $\epsilon_k^{\text{HF}}/\epsilon_F^0 = x^2 - 0.663 r_s F(x)$; band width $1 + 0.33 r_s$, too large
- $F'(1) = -\infty$: $m^* \rightarrow 0$ like $1/\ln$, no density of states at k_F — unscreened $1/q^2$, cured by RPA
- $E_0/N = 2.21/r_s^2 - 0.916/r_s$ Ry: exchange alone binds a metal at $r_s = 4.83$; correlation (-0.06 Ry) is comparable

Friday: the density as the variable

- $\hat{T}$, $\hat{V}$ are the same for all matter; only v_{ext} differs
- The energy needs only $\rho^{(1)}, \rho^{(2)}$; for a determinant $\rho^{(2)}$ factorises — the boundary of mean field
- HK I: n_0 fixes v_{ext} (four lines, strict variational principle). HK II: a universal $F[n]$, $E_0 = \min E[n]$
- Existence only; v -representability (Levy-Lieb); ground states only
- $\delta E/\delta n = \mu$; Thomas-Fermi from the gas: cheap, 1% wrong in the largest term, no shells

Next week, and a closing remark

Week 43, October 22–23: density functional theory, part II

Thursday: the Kohn-Sham construction (section 8.5): a fictitious non-interacting system with the *same density*, the orbitals back in for the kinetic energy, the Kohn-Sham equations as a Hartree-Fock-like self-consistent problem with a *local* potential $v_{xc}(r) = \delta E_{xc} / \delta n$.

Friday: the local density approximation (section 8.6), derived rather than quoted — the exchange hole of the Fermi sea, $-0.916/r_s$ recovered from $\rho^{(1)}$, Ceperley-Alder correlation — and Kohn-Sham in practice (section 8.7): the same trap system as week 38 solved by Hartree-Fock, Kohn-Sham and exactly, self-interaction, and what the Kohn-Sham eigenvalues mean.

Reading: chapter 8, sections 8.5–8.7. Code: `dft.py`. No exercise session in week 43 either: first midterm.

Two bargains in one week. Thursday's determinant got the mechanism of metallic cohesion right and the Fermi surface wrong, for one reason: an unscreened $1/q^2$. Friday replaced the wave function by the density, exactly, and then had to admit that the exact functional is unknown and the first guess at it, from Thursday's gas, misses the shell structure. Next week the two threads meet: orbitals for the kinetic energy, the electron gas for everything else.

The best of both: Kohn and Sham, 1965.